

LAG-2 Phase 2 v0.1 outline — S_geom reduction integral $D_{IV}^5 \rightarrow \mathbb{R}^{\{3,1\}}$

Lyra

2026-05-19

LAG-2 Phase 2 v0.1 — S_geom Reduction Integral

Status

Phase 2 structural identification is CLOSED at I-tier per T2343 (LAG-2 Phase 2.2+2.3, Monday 2026-05-17, with K-audit refinement Monday 2026-05-18). The structural form of S_geom reduction from $D_{IV}^5 \rightarrow \mathbb{R}^{\{3,1\}}$ is identified; the explicit Faraut-Koranyi integration is the multi-month Phase 2 v0.2-v0.4 work.

Phase 2 sub-tasks (multi-month staged)

Phase 2.1 — Cartan-Wolf canonicity (DONE, T2342)

$H^4 \subset M(D_{IV}^5)$ embedding via Cartan-Wolf classification of totally-geodesic sub-manifolds of Hermitian symmetric spaces. Classical Helgason 1978 + Wolf 1984. T2342 D-tier filed Monday.

Phase 2.2 — S_geom reduction integral structural identification (DONE, T2343 I-tier)

Reduction integral form:

$$S_{\text{geom}}_{4D} = \int_{\{\mathbb{R}^{\{3,1\}}\}} L_{EH}(x) d^4x = \int_{\{H^4 \subset M\}} [\text{reduction map}](L_{BST}(z, \bar{z})) d\text{vol}_B(z)$$

where: - L_{BST} = Bergman-Kähler-Einstein action functional on D_{IV}^5 - reduction map = Faraut-Koranyi boundary integration over internal 6-dim complement - L_{EH} = standard Einstein-Hilbert action in 4D

T2343 filed Monday at I-tier structural identification — the reduction structural form is identified; the explicit integration is multi-month.

Phase 2.3 — Faraut-Koranyi internal-complement integration (OPEN, multi-week)

The 6-dim internal-complement integration:

$$\int_{\{\text{Internal}^6\}} [\text{Bergman-volume measure}] = \pi^{n_C} \cdot (\Gamma\text{-factor product})$$

For type-IV D_{IV}^5 : Faraut-Koranyi 1990 gives $\text{Vol}_6(\text{internal}) \propto \pi^{n_C - \text{rank}^2} \Gamma(g/2) \Gamma(g/2-1) \dots$ up to BST primary normalization.

Multi-week derivation steps: - (a) Explicit Hua coordinate decomposition of D_{IV}^5 as $H^4 \times \text{Internal}^6$ (~1-2 wk) - (b) Bergman volume integration over Internal^6 with Faraut-Koranyi Γ -factor product (~2-3 wk) - (c) BST primary identification of the integrated coefficient (~1 wk after Faraut-Koranyi closure) - (d) Numerical match against measured G_{Newton} and $\Lambda_{\text{cosmological}}$ (~1-2 wk)

Phase 2.4 — Einstein-Hilbert recovery from reduced action (OPEN, multi-week)

Once the reduction integral is closed (Phase 2.3), the 4D effective action:

$$S_{\text{eff}_4D} = (\text{Faraut-Koranyi integrated coefficient}) \cdot \int_{\mathbb{R}^{\{3,1\}}} R(x) \sqrt{(-g)} d^4x$$

should recover standard Einstein-Hilbert with prefactor $1/(16\pi G)$. The BST-primary form of (Faraut-Koranyi integrated coefficient) gives G derivation per Paper #120 v0.2 structural-prediction framework.

Multi-week derivation: explicit computation of the Faraut-Koranyi prefactor in BST primaries; comparison to G_{Newton} .

Connection to other LAG/Möbius work

Connection	Description
Phase 1 (T2340/T2342) LAG-1 Session 10 Step 5.2 (T2379 prep doc)	DONE — 4+6 split + Cartan-Wolf canonicity Step 5.2 Faraut-Koranyi volume framework SHARED with Phase 2.3
Gap #3 t^* identification (T2367)	Saddle parameter $t^* = 5/968$ derived from heat kernel — connects to Faraut-Koranyi via volume integration
Paper #120 v0.2 G/inertia paper	Substrate-mediated G ontology relies on Phase 2 closure for numerical G derivation

The Phase 2.3 + 2.4 substantive derivation is the multi-week bottleneck for closing both: - Paper #120 v0.2 Section 4 (G from BST primaries, currently I-tier) - LAG-1 Session 10 Step 5.5 (ind(D) BST primary value, currently candidate set {13, 15}) - LAG-2 Phase 3+ (S_BST term-by-term reductions, currently I-tier T2344)

Honest scoping per Cal External_Survivability_Checklist

Closed: Phase 2 structural identification at I-tier (T2343, Monday). This v0.1 outline confirms scope and identifies Phase 2.3 + 2.4 sub-tasks.

Open multi-week: explicit Faraut-Koranyi internal-complement integration + BST primary identification of integrated coefficient + Einstein-Hilbert recovery numerical match.

Per Cal Coincidence_Filter_Risk: NOT “LAG-2 Phase 2 closed at D-tier.” Correct: “Phase 2 structural identification at I-tier (T2343); explicit reduction integral multi-week per Phase 2.3 + 2.4.”

Total LAG-2 horizon (per Paper #115 v0.5 Section 9.x)

Phase	Status	Scope
1 (4+6 split)	CLOSED structural (T2340 + T2342)	DONE
2 (S_geom reduction)	I-tier structural (T2343); v0.2-v0.4 substantive	Multi-month
3 (S_BST 6 terms)	I-tier prefactor identification (T2344); numerical end-to-end	Multi-month
4 (Einstein eq emergence)	D-tier corollary of T2343 (T2345); operator-level	3-6 mo
5 (SM gauge)	D-tier structural integer ID (T2346); kinetic terms	6+ mo

Total LAG-2 horizon: ~year of focused work to D-tier closure across all phases. Consistent with Casey’s original “multi-year, closes BST as physical theory” framing.

Filing notes

Status: v0.1 outline per Casey “finish your board” Tuesday directive. Companion to Phase 1 outline. Phase 2 was substantially closed Monday (T2343 structural); this outline document confirms multi-month sub-task scope.

Owner: Lyra primary.

Next: v0.2 Phase 2.3 Faraut-Koranyi internal-complement integration explicit (~2-3 wk substantive). When Lyra has bandwidth for multi-week LAG-2 substantive work.

— Lyra, LAG-2 Phase 2 v0.1 outline filed per Casey “finish your board,” 2026-05-19 ~11:20 EDT.